

Nonrandom X inactivation and selection of fragile X full mutation in fetal fibroblasts.

PMID: 10449653 | Indexed in PubMed

In the fragile X female carriers the degree of cognitive impairment appears to be correlated with activation status of the X chromosome bearing the expanded trinucleotide repeat in the promoter of the FMR1 gene. In this study we asked if the deviations from the primarily random pattern of X inactivation are related to the selection which is thought to occur against cells carrying the fragile X full mutation (FM) on the active X chromosome. A fibroblast culture derived from a 20-week FM female fetus was serially passaged. The activation ratio (AR) of the culture increased from 0.68 to 0.92 between passages 2 and 9. All higher passage cells (up to 34 passages) display an AR of 1.0, indicating complete absence of cells in which the normal X chromosome would be inactivated. Of 29 clones established from the fetal culture with AR of 0.8, 28 had no visible 5.2-kb band on Southern blots indicating that these 28 clones consisted entirely of cells with FM on their inactive X chromosome. Only a single clone carried the FM on its active X chromosome. The figure of 1 of 29 is much lower than our expectation based on the AR of mass culture. Therefore cloning and serial cultivation indicate the possibility of selection depending on the activation status of the expanded X chromosome in fetal FM female fibroblasts.

Neuropsychology and brain morphology in Klinefelter syndrome - the impact of genetics.

PMID: 24865607 | Indexed in PubMed

Klinefelter syndrome (KS, 47,XXY) is associated with increased psychiatric morbidity and cognitive disabilities, although the neuropsychological phenotype shows great variability. Androgen receptor polymorphism (CAG repeat length), skewed X-chromosome inactivation and parent-of-origin of the extra X-chromosome have been suggested to influence cognitive function and psychological traits. These issues have not been clarified for KS patients. We studied X-chromosome inactivation pattern, CAG repeat length and parent-of-origin in relation to educational and cohabitation status, personality and autism traits, psychological distress, cognitive function and brain volumes in 73 KS patients and 73 controls. Grey matter (GM) volume of left insula was significantly decreased in KS patients with skewed X-inactivation ($z = 5.78$) and we observed a borderline significant difference in global brain matter volume where KS patients with skewed X-chromosome inactivation tended to have smaller brains. Skewed X-inactivation, CAG repeat length and parent-of-origin were not correlated with educational and marital status, personality traits, autism traits, and psychological distress, prevalence of depression and anxiety or cognitive function. Interestingly our results regarding brain volumes indicate that X-inactivation has an influence on GM volume in left insula and might also be related to global GM volume, indicating a possible effect of X-linked genes on the development of GM volume in KS patient. Skewed X-inactivation, CAG repeat length and parent-of-origin have no impact on the neuropsychological phenotype in KS (<http://www.clinicaltrials.gov> (Clinical trial NCT00999310)).

Mining and analysis of adverse drug reactions associated with perampanel based on FAERS database.

PMID: 37315406 | Indexed in PubMed

Perampanel (PER) is a non-competitive AMPA glutamate receptor antagonist used as an anti-seizure medication. Large post-marketing databases are still lacking for safety analysis of the new generation of anti-seizure medications. Based on the FDA's adverse event reporting system (FAERS) database, this study aimed to investigate, assess, and offer evidence for the safety of PER to support clinical decision-making. Perampanel-related adverse reaction signals were mined using the reporting odds ratio (ROR), medicines and healthcare products regulatory agency (MHRA), and Bayesian confidence propagation neural network (BCPNN). The rate and occurrence of reported adverse responses were examined. With the three methodologies used in combination, 83 signals mostly related to psychosis and different nervous system disorders were detected. Among them, suicide behavior, respiratory depression, hepatotoxicity, cognitive impairment, and other possible novel signals warranted consideration. Further examination of the age and gender differences in the detected signals revealed that elderly patients should be closely monitored for any change in consciousness and the occurrence of movement disorders; male patients should be observed for negative mental reactions like a personal attack and homicidal ideation; and female patients should be watched for the occurrence of negative reactions in memory, weight, vision, liver function, and other specific areas. This study found that PER had the risk of causing suicide behavior, respiratory depression, hepatotoxicity, and cognitive impairment among other adverse effects. When used clinically, PER should be closely monitored for the occurrence of adverse effects on mental health and behavior. However, these results should be interpreted with caution.

Perampanel with concomitant levetiracetam and topiramate: Post hoc analysis of adverse events related to hostility and aggression.

PMID: 28830031 | Indexed in PubMed

In 4 Phase III registration trials (3 in patients with partial seizures, N=1480; 1 in patients with PGTCs, N=163), perampanel administered to patients already receiving 1-3 concomitant antiepileptic drugs (AEDs) demonstrated statistically superior efficacy compared to placebo in reducing seizure frequency. However, use of perampanel in these studies was associated with a risk of psychiatric and behavioral adverse reactions, including aggression, hostility, irritability, anger, and homicidal ideation and threats. The present study is a post hoc analysis of pooled data from these 4 trials to determine if concomitant treatment with levetiracetam and/or topiramate increased the risk of hostility- and aggression-related AEs. Treatment-emergent AEs (TEAEs) were determined using a "Narrow & Broad" search based on the Medical Dictionary for Regulatory Activities (MedDRA) standard MedDRA query (SMQ) for hostility- and aggression-related events. The rate of hostility- and aggression-related TEAEs was observed to be similar among perampanel-treated patients: a) receiving levetiracetam (N=340) compared to those not receiving levetiracetam (N=779); b) receiving topiramate (N=223) compared to those not receiving topiramate (N=896); and c) receiving both levetiracetam and topiramate (N=47) compared to those not receiving levetiracetam and topiramate (N=1072). Severe and serious TEAEs related to hostility and aggression were rare and occurred at a similar rate regardless of concomitant levetiracetam and/or topiramate therapy. Taken together, these results suggest that concomitant treatment with levetiracetam and/or topiramate has no appreciable effect on the occurrence of hostility- or aggression-related TEAEs in patients receiving perampanel.

Improved ambient air quality is associated with decreased prevalence of childhood

asthma and infancy shortly after weaning is a sensitive exposure window.

PMID: 37458141 | Indexed in PubMed

The urban ambient air quality has been largely improved in the past decade. It is unknown whether childhood asthma prevalence is still increasing in ever top-ranking city of Shanghai, whether the improved air quality is beneficial for children's asthma and what time window of exposure plays critical roles. Using a repeat cross-sectional design, we analyzed the association between early life exposure to particles and wheezing/asthma in each individual and combined surveys in 2011 and 2019, respectively, in 11,825 preschool children in Shanghai. A significantly lower prevalence of doctor-diagnosed asthma (DDA) (6.6% vs. 10.5%, $p < 0.001$) and wheezing (10.5% vs. 23.2%, $p < 0.001$) was observed in 2019 compared to 2011. Exposure to fine particulate matter (PM_{2.5}), coarse particles (PM_{2.5-10}) and inhalable particles (PM₁₀) was decreased in 2019 by 6.3%, 35.4%, and 44.7% in uterus and 24.3%, 20.2%, and 31.8% in infancy, respectively. Multilevel log-binomial regression analysis showed exposure in infancy had independent association with wheezing/DDA adjusting for exposure in uterus. For each interquartile range (IQR) increase of infancy PM_{2.5}, PM_{2.5-10} and PM₁₀ exposure, the odds ratios were 1.39 (95% confidence interval (CI): 1.24-1.56), 1.51 (95% CI:1.15-1.98) and 1.53 (95% CI:1.27-1.85) for DDA, respectively. The distributed lag non-linear model showed the sensitive exposure window (SEW) was 5.5-11 months after birth. Stratified analysis showed the SEWs were at or shortly after weaning, but only in those with <6 months of exclusive breastfeeding. Improved ambient PM benefits in decreasing childhood asthma prevalence. We firstly reported the finding of SEW to PM at or closely after weaning on childhood asthma.

Urban environment during pregnancy and lung function, wheezing, and asthma in school-age children. The generation R study.

PMID: 38219897 | Indexed in PubMed

The urban environment during pregnancy may influence child's respiratory health, but scarce evidence exists on systematic evaluation of multiple urban exposures (e.g., air pollution, natural spaces, noise, built environment) on children's lung function, wheezing, and asthma development. We aimed to examine the association of the urban environment during pregnancy with lung function, preschool wheezing, and school-age asthma. We included 5624 mother-child pairs participating in a population-based prospective birth cohort. We estimated 30 urban environmental exposures including air pollution, road traffic noise, traffic, green spaces, blue spaces, and built environment during pregnancy. At 10 years of age, lung function was measured by spirometry. Information on preschool wheezing and physician-diagnosed school-age asthma was obtained from multiple questionnaires. We described single-exposure associations with respiratory outcomes using an exposome-wide association study. We also identified patterns of urban exposures with hierarchical clustering on principal components analysis and examined their associations with respiratory outcomes using multivariate regression models. Single-exposure analyses showed associations of higher particulate matter (PM) with lower mid-expiratory flow (FEF_{25-75%}) (e.g., for PM < 2.5 μ m of diameter [PM_{2.5}] z-score = -0.06 [-0.09, -0.03]) and higher forced expiratory volume in 1s (FEV₁) and forced vital capacity (FVC) (e.g., for PM_{2.5} FEV₁ 0.05 [0.02, 0.08]) after correction for multiple-hypothesis testing. Cluster analysis described three patterns of urban exposures during pregnancy and showed that the cluster characterised by higher levels of air pollution, noise, walkability, street connectivity, and lower levels of natural spaces were associated with lower FEF_{25-75%} (-0.08 [-0.17, 0.00]), and higher odds of preschool wheezing (1.21 [1.03, 1.43]). This study shows that the characteristics of the urban environment during pregnancy are of relevance to

the offspring's respiratory health during childhood.

Early-Life Exposure to Air Pollution and Childhood Asthma Cumulative Incidence in the ECHO CREW Consortium.

PMID: 38416497 | Indexed in PubMed

Exposure to outdoor air pollution contributes to childhood asthma development, but many studies lack the geographic, racial and ethnic, and socioeconomic diversity to evaluate susceptibility by individual-level and community-level contextual factors. To examine early life exposure to fine particulate matter (PM_{2.5}) and nitrogen oxide (NO₂) air pollution and asthma risk by early and middle childhood, and whether individual and community-level characteristics modify associations between air pollution exposure and asthma. This cohort study included children enrolled in cohorts participating in the Children's Respiratory and Environmental Workgroup consortium. The birth cohorts were located throughout the US, recruited between 1987 and 2007, and followed up through age 11 years. The survival analysis was adjusted for mother's education, parental asthma, smoking during pregnancy, child's race and ethnicity, sex, neighborhood characteristics, and cohort. Statistical analysis was performed from February 2022 to December 2023. Early-life exposures to PM_{2.5} and NO₂ according to participants' birth address. Caregiver report of physician-diagnosed asthma through early (age 4 years) and middle (age 11 years) childhood. Among 5279 children included, 1659 (31.4%) were Black, 835 (15.8%) were Hispanic, 2555 (48.4%) were White, and 229 (4.3%) were other race or ethnicity; 2721 (51.5%) were male and 2596 (49.2%) were female; 1305 children (24.7%) had asthma by 11 years of age and 954 (18.1%) had asthma by 4 years of age. Mean values of pollutants over the first 3 years of life were associated with asthma incidence. A 1 IQR increase in NO₂ (6.1 µg/m³) was associated with increased asthma incidence among children younger than 5 years (HR, 1.25 [95% CI, 1.03-1.52]) and children younger than 11 years (HR, 1.22 [95% CI, 1.04-1.44]). A 1 IQR increase in PM_{2.5} (3.4 µg/m³) was associated with increased asthma incidence among children younger than 5 years (HR, 1.31 [95% CI, 1.04-1.66]) and children younger than 11 years (OR, 1.23 [95% CI, 1.01-1.50]). Associations of PM_{2.5} or NO₂ with asthma were increased when mothers had less than a high school diploma, among Black children, in communities with fewer child opportunities, and in census tracts with higher percentage Black population and population density; for example, there was a significantly higher association between PM_{2.5} and asthma incidence by younger than 5 years of age in Black children (HR, 1.60 [95% CI, 1.15-2.22]) compared with White children (HR, 1.17 [95% CI, 0.90-1.52]). In this cohort study, early life air pollution was associated with increased asthma incidence by early and middle childhood, with higher risk among minoritized families living in urban communities characterized by fewer opportunities and resources and multiple environmental coexposures. Reducing asthma risk in the US requires air pollution regulation and reduction combined with greater environmental, educational, and health equity at the community level.

Improved childhood asthma control after exposure reduction interventions for desert dust and anthropogenic air pollution: the MEDEA randomised controlled trial.

PMID: 38388489 | Indexed in PubMed

Elevated particulate matter (PM) concentrations of anthropogenic and/or desert dust origin are

associated with increased morbidity among children with asthma. The Mitigating the Health Effects of Desert Dust Storms Using Exposure-Reduction Approaches randomised controlled trial assessed the impact of exposure reduction recommendations, including indoor air filtration, on childhood asthma control during high desert dust storms (DDS) season in Cyprus and Greece. Primary school children with asthma were randomised into three parallel groups: (a) no intervention (controls); (b) outdoor intervention (early alerts notifications, recommendations to stay indoors and limit outdoor physical activity during DDS) and (c) combined intervention (same as (b) combined with indoor air purification with high efficiency particulate air filters in children's homes and school classrooms. Asthma symptom control was assessed using the childhood Asthma Control Test (c-ACT), spirometry (forced expiratory volume in 1 s (FEV1), forced vital capacity (FVC)) and fractional exhaled nitric oxide (FeNO). In total, 182 children with asthma (age; mean=9.5, SD=1.63) were evaluated during 2019 and 2021. After three follow-up months, the combined intervention group demonstrated a significant improvement in c-ACT in comparison to controls ($\beta=2.63$, 95% CI 0.72 to 4.54, $p=0.007$), which was more profound among atopic children ($\beta=3.56$, 95% CI 0.04 to 7.07, $p=0.047$). Similarly, FEV1% predicted ($\beta=4.26$, 95% CI 0.54 to 7.99, $p=0.025$), the need for any asthma medication and unscheduled clinician visits, but not FVC% and FeNO, were significantly improved in the combined intervention compared with controls. Recommendations to reduce exposure and use of indoor air filtration in areas with high PM pollution may improve symptom control and lung function in children with asthma. NCT03503812.

Machine learning-driven identification of air toxic combinations associated with asthma symptoms among elementary school children in Spokane, Washington, USA.

PMID: 38387571 | Indexed in PubMed

Air toxics are atmospheric pollutants with hazardous effects on health and the environment. Although methodological constraints have limited the number of air toxics assessed for associations with health and disease, advances in machine learning (ML) enable the assessment of a much larger set of environmental exposures. We used ML methods to conduct a retrospective study to identify combinations of 109 air toxics associated with asthma symptoms among 269 elementary school students in Spokane, Washington. Data on the frequency of asthma symptoms for these children were obtained from Spokane Public Schools. Their exposure to air toxics was estimated by using the Environmental Protection Agency's Air Toxics Screening Assessment and National Air Toxics Assessment. We defined three exposure periods: the most recent year (2019), the last three years (2017-2019), and the last five years (2014-2019). We analyzed the data using the ML-based Data-driven ExposurE Profile (DEEP) extraction method. DEEP identified 25 air toxic combinations associated with asthma symptoms in at least one exposure period. Three combinations (1,1,1-trichloroethane, 2-nitropropane, and 2,4,6-trichlorophenol) were significantly associated with asthma symptoms in all three exposure periods. Four air toxics (1,1,1-trichloroethane, 1,1,2,2-tetrachloroethane, BIS (2-ethylhexyl) phthalate (DEHP), and 2,4-dinitrophenol) were associated only in combination with other toxics, and would not have been identified by traditional statistical methods. The application of DEEP also identified a vulnerable subpopulation of children who were exposed to 13 of the 25 significant combinations in at least one exposure period. On average, these children experienced the largest number of asthma symptoms in our sample. By providing evidence on air toxic combinations associated with childhood asthma, our findings may contribute to the regulation of these toxics to improve children's respiratory health.

Air pollution and childhood asthma: recent advances and future directions.

PMID: 19663041 | Indexed in PubMed

Current levels of air pollution are consistently associated with asthma development and morbidity among children, suggesting that current regulatory policies may be insufficient. This review will describe recent studies that have examined specific emission sources or components of pollutants that may be associated with pediatric asthma and identify subpopulations that may be particularly susceptible to the effects of air pollution exposure. Important advances include new characterizations of the effects of traffic-related air pollution in urban areas. They also include the application of novel exposure and outcome measures such as pollution estimates derived from land use regression modeling and biological markers of airway inflammation. Additionally, studies have identified host susceptibility characteristics that may modify responses to air pollution exposure, including polymorphisms in oxidative stress genes and epigenetic alterations. Identifying specific sources and toxic constituents of air pollution and accurately assessing air pollutant-related asthma outcomes are needed to better direct control strategies. Further research is needed to identify additional host factors that confer increased susceptibility to air pollution exposure. Future therapy to reduce the adverse effects of air pollution on respiratory disease will likely depend on targeting susceptible populations for intervention.
